

MIAE 221-Midterm A-2025 Exam Solutions and Key Concepts

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MIAE 221 - Materials Science - Midterm 2025**Select the best answer on the Scantron IBM Sheet. USE A PENCIL!!!**

Write the version of your exam (A, B or C) on the Scantron Sheet.

Questions 1-20 are worth two marks each; Questions 21-30 are one mark each.

Hand-in only the Scantron MC answer sheet – keep the question paper.

Version

A

1. The crystal structure illustrated in Figure 1 is:

- (a) BCC
- (b) HCP
- (c) FCC
- (d) SC (Simple Cubic)
- (e) OLP (Orthorhombic Loose Packed)

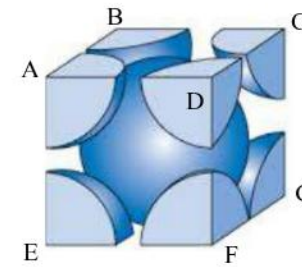


Figure 1

2. In Figure 1, which of the following is a close-packed plane?

- (a) ABCD
- (b) ABGF
- (c) ACGE
- (d) BGE
- (e) There are no close-packed planes in this structure.

3. In Figure 1, the planes ABGF and BCFE, are all members of this family of planes:

- (a) {100}
- (b) {110}
- (c) {111}
- (d) <110>
- (e) <100>

4. In Figure 1, the planar density of (110) plane, in terms of atomic radius, R is:

- (a) $1/\sqrt{2}R^2$
- (b) $1/2\sqrt{2}R^2$
- (c) $1/4\sqrt{2}R^2$
- (d) $1/8\sqrt{2}R^2$
- (e) $3/8\sqrt{2}R^2$

5. Which of the following mechanical properties **cannot** be measured from a tensile test?

- (a) Hardness
- (b) Yield strength
- (c) Tensile strength
- (d) Young's Modulus
- (e) Ductility

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Version A

6. Which of the following electron configurations is for an inert gas?

(a) $1s^2 2s^2 2p^6 3s^2 3p^6$

✓ This is a noble gas configuration (Argon, Z = 18).

(b) $1s^2 2s^2 2p^6 3s^2$

(c) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$

(d) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^2 4s^2$

(e) $1s^2 2s^2 2p^6 3s^1$

7. Hydrogen gas can be purified by diffusion through a palladium sheet. In this system, calculate the number of kilograms of hydrogen that pass per hour through an 8-mm-thick sheet of palladium having an area of 1.20 m^2 at 500°C . Assume a diffusion coefficient of $1.0 \times 10^{-8} \text{ m}^2/\text{s}$ at this temperature, that the concentrations at the high- and low-pressure sides of the plate are 2.4 and 0.6 kg of hydrogen per cubic meter of palladium respectively, and that steady-state conditions have been attained.

(a) 2.7×10^{-6}

(b) 2.16×10^{-3}

(c) 7.5×10^{-10}

(d) 9.72×10^{-3}

(e) 1.62×10^{-4}

$$JA = -D \frac{C_2 - C_1}{\Delta x} A = 10^{-8} \frac{2.4 - 0.6}{8 \times 10^{-3}} \times 1.2 = 2.7 \times 10^{-6} \text{ kg/s} = 9.72 \times 10^{-3} \text{ kg/h}$$

8. Tungsten has a BCC crystal structure, an atomic radius of 0.137 nm and an atomic weight of 183.85 g/mol. The theoretical density of tungsten is closest to:

(a) 1.93 g/cm³

(b) 5.18 g/cm³

(c) 9.65 g/cm³

(d) 19.3 g/cm³

(e) 38.6 g/cm³

$$\rho = \frac{n A_w}{V_c N_A} = \frac{2 \times 183.85}{3.167 \times 10^{-21} \times 6.022 \times 10^{23}} = 19.28 \text{ g/cm}^3$$

$$a = \frac{4R}{\sqrt{3}} = \frac{4 \times 0.137}{\sqrt{3}} = 0.3164 \text{ nm}$$

$$V_c = a^3 = (0.3164 \times 10^{-9})^3 = 3.167 \times 10^{-27} \text{ m}^3 = 3.167 \times 10^{-21} \text{ cm}^3$$

9. Determine the Miller indices for the planes shown in the unit cell presented in Figure 2:

(a) A is (232), B is (112)

(b) A is (320), B is (002)

(c) A is [320], B is [112]

(d) A is (6 $\bar{3}$ 3), B is (212)

(e) A is (3 $\bar{2}$ 4), B is (221)

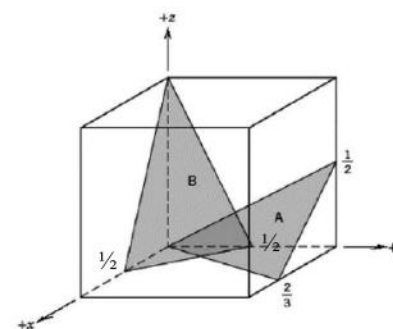


Figure 2

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10. 15% nickel can be added to copper in the molten state and cooled to give a substitutional solid solution. This means that the nickel atoms are:

a) is that of nickel

b) is that of copper with the nickel atoms squeezed into the spaces between the copper atoms.

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10. 15% nickel can be added to copper in the molten state and cooled to give a substitutional solid solution. This means that the crystal structure of the solid solution:

- a) is that of nickel with the copper atoms squeezed into the spaces between the nickel atoms.
- b) is that of copper with the nickel atoms squeezed into the spaces between the copper atoms.
- c) consists of grains of pure copper which make up 85% of the weight of the material. The remaining 15% occurs as grains of pure nickel.
- d) is basically that of FCC copper with some of the copper atoms replaced by nickel atoms.
- e) consists of unit cells of FCC copper equally interspersed with unit cells of FCC nickel.

11. The difference between adjacent grains in a polycrystalline pure metal is one of:

- (a) Density
- (b) Lattice parameter
- (c) Crystal structure
- (d) Composition
- (e) Orientation

12. The electronegativities of Sodium and Chlorine are 1.0 and 2.9 respectively; this indicates that the bonding between these elements is predominantly:

- (a) Metallic.
- (b) Covalent.
- (c) Ionic.
- (d) Van der Waals.
- (e) Hydrogen bonding.

13. With regard to defects, the following statements is false?

- (a) The concentration of defects is a function of temperature.
- (b) Grain boundaries are defects.
- (c) Pores (bulk defects) can be formed during solidification of molten metal.
- (d) Impurity atoms can take up either substitutional or interstitial positions in the crystal structure depending on their size.
- (e) Vacancies are required for substitutional solid state diffusion.

14. The defect shown in the adjacent Figure is a representation of:

- (a) a vacancy.
- (b) a screw dislocation.
- (c) an edge dislocation.
- (d) a substitutional impurity.
- (e) a grain boundary.

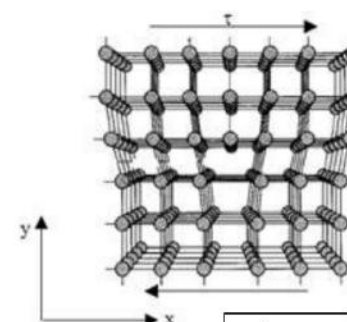


Figure 3

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15. The Atomic Packing Factors (APF) for the BCC : FCC : HCP crystal systems are respectively:

- (a) 0.74 : 0.74 : 0.68
- (b) 0.68 : 0.74 : 0.74
- (c) 0.74 : 0.68 : 0.68
- (d) 0.68 : 0.74 : 0.68
- (e) None of the above choices.

16. Which of the following best describes a body centered tetragonal unit cell?

- (a) $a = b \neq c$, $\alpha = \beta = \gamma = 90^\circ$ with an atom at each corner plus one at the centre.
- (b) $a = b = c$, $\alpha = \beta = \gamma = 90^\circ$ with an atom at each corner plus one at the centre.
- (c) $a \neq b \neq c$, $\alpha = \beta = \gamma = 90^\circ$ with an atom at each corner plus one at the centre.
- (d) $a = b = c$, $\alpha = \beta = \gamma = 120^\circ$ with an atom at each corner plus one at the centre.
- (e) $a = b = c$, $\alpha = \beta = \gamma = 90^\circ$ with an atom at each corner plus one at the centre.



15. The Atomic Packing Factors (APF) for the BCC : FCC : HCP crystal systems are respectively:

- (a) 0.74 : 0.74 : 0.68
- (b) 0.68 : 0.74 : 0.74
- (c) 0.74 : 0.68 : 0.68
- (d) 0.68 : 0.74 : 0.68
- (e) None of the above choices.

16. Which of the following best describes a body centered **tetragonal** unit cell?

- (a) $a = b \neq c$, $\alpha = \beta = \gamma = 90^\circ$ with an atom at each corner and one in the centre.
- (b) $a = b = c$, $\alpha = \beta = \gamma = 90^\circ$ with an atom at each corner plus one at the centre.
- (c) $a \neq b \neq c$, $\alpha = \beta = \gamma = 90^\circ$ with an atom at each corner plus one at the centre.
- (d) $a = b \neq c$, $\alpha = \beta = \gamma = 90^\circ$ with an atom at each corner and one in the centre of each face.
- (e) $a \neq b = c$, $\alpha = \beta = \gamma = 90^\circ$ with an atom at each corner plus one at the centre.

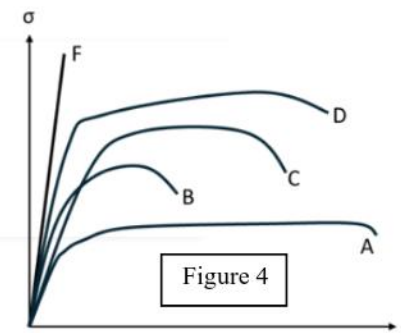
17. At 660°C the fractional concentration of vacancies in aluminum is 9×10^{-5} . The energy for vacancy formation is therefore:

- (a) 0.72 eV
- (b) 0.68 eV
- (c) 0.75 eV
- (d) 0.79 eV
- (e) None of the above choices.

$$\frac{N_v}{N} = \exp\left(\frac{-Q_v}{kT}\right) \Rightarrow 9 \times 10^{-5} = \exp\left(\frac{-Q_v}{8.62 \times 10^{-5} \times (660 + 273)}\right)$$
$$\Rightarrow \ln(9 \times 10^{-5}) = -\frac{Q_v}{0.0804} \Rightarrow Q_v = 0.749 \text{ eV}$$

18. In Figure 4, the tensile curve for five hypothetical materials is illustrated. Which material is the stiffest?

- (a) A
- (b) B
- (c) C
- (d) D
- (e) F



19. In Figure Y, which material has the highest yield strength?

- (a) A
- (b) B
- (c) C
- (d) D
- (e) F

20. In Figure Y, which material shows the highest ductility?

- (a) A
- (b) B
- (c) C
- (d) D
- (e) F



True or False: For Questions 21 – 30, decide whether the following statements are True or False and shade in the appropriate box on the Scantron Answer sheet. Each correct answer is worth 1 mark. For **True**, shade in letter (a). For **False** shade in letter (b).

21. Allotropy or polymorphism is a property exhibited by some elements which means that they crystallize in different crystal structures at different ranges of temperature and pressure. **T**
22. Non-crystalline (amorphous) materials have grain boundaries, but do not have crystals. **F**
23. Krypton (electronic structure: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$) is a highly reactive gas. **F**
24. In the cubic crystal system, the direction perpendicular to the (hkl) plane is the [hkl] direction. **T**
25. The number of atoms, n, contained within a BCC unit cell is 9 as there is one atom at each corner of the cube and one in the centre. **F**
26. Covalent bonding is a form of primary bonding which is based on permanent or induced (temporary) dipoles between species. **F**
27. The reason why ice floats on water is due to the lower density of the ice structure which is a result of the secondary (hydrogen) bonding that forms between H₂O molecules during freezing. **T**
28. Carbon has the e ion of the 2s and 2p

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- electrons it can form four equal bonds with four hydrogen atoms to form a methane molecule. **T**
29. The expected diffraction angle, (2θ), is 40.69° for the first-order reflection from the (113) set of planes for FCC platinum (a = 0.3923 nm) when monochromatic radiation of wavelength 0.1542 nm is used. **F**
- $$n\lambda = 2d_{hkl} \sin \theta$$
$$\Rightarrow \sin \theta = \frac{1 \times 0.1542}{2 \times 0.1183} = 0.652 \Rightarrow \theta = 40.67^\circ \Rightarrow \boxed{2\theta = 81.34^\circ}$$
$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} = \frac{0.3923}{\sqrt{1+1+9}} = 0.1183 \text{ nm}$$
30. Crystalline atomic structures have both short-range and long-range order whereas amorphous or non-crystalline materials (such as glass) have only short-range order. **F**

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Equation Sheet

- $A \cong Z + N$
- $\%ionicity = \{1 - e^{\frac{4R}{\sqrt{3}}}\}$
- $E_{total} = E_A + E_R$
- $a = \frac{4R}{\sqrt{3}}$

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Version A

MIAE 221
Equation Sheet

$A \cong Z + N$

$F_N = F_A + F_R$

$E_N = \int_r^\infty F_N dr = E_A + E_R$

$\%ionicity = \{1 - \exp[-(0.25)(X_A - X_B)^2]\} \times 100$

$a = 2R\sqrt{2}$

$a = \frac{4R}{\sqrt{3}}$

$APF = \frac{\text{volume of atoms in unit cell}}{\text{total unit cell volume}}$

$\rho = \frac{nA}{V_C N_A}$

$LD = \frac{n_a}{L_t}$

$PD = \frac{n_a}{A_p}$

$n = 2^{G-1}$

$n\lambda = 2d_{hkl} \sin \theta$

$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$

$N_v = N \exp\left(-\frac{Q_v}{kT}\right)$

$C_1 = \frac{m_1}{m_1 + m_2} \times 100 \quad J = \frac{M}{At}$

$J = -D \frac{dC}{dx}$

$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2}$

$\frac{C_x - C_0}{C_s - C_0} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$

$D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$

$\sigma_{Eng} = \frac{F}{A_0}$

$\epsilon_{Eng} = \frac{\Delta l}{l_0}$

$\sigma = E\epsilon$

$\tau = G\gamma$

$\nu = -\frac{\epsilon_x}{\epsilon_z} = -\frac{\epsilon_y}{\epsilon_z}$

$E = 2G(1 + \nu)$

$\%El = \left(\frac{l_f - l_0}{l_0}\right) \times 100$

$U_r = \frac{1}{2} \sigma_y \epsilon_y = \frac{\sigma_y^2}{2E}$

Constants

$N_A = 6.022 \times 10^{23} \text{ atoms/mol}$

$k = 8.62 \times 10^{-5} \text{ eV/atom.K or } 1.38 \times 10^{-23} \text{ J/atom.K}$

$R = 8.314 \text{ J/mol}$

z	erf(z)	z	erf(z)	z	erf(z)
0	0	0.55	0.5633	1.3	0.9340
0.025	0.0282	0.60	0.6039	1.4	0.9523
0.05	0.0564	0.65	0.6420	1.5	0.9661
0.10	0.1125	0.70	0.6778	1.6	0.9763
0.15	0.1680	0.75	0.7112	1.7	0.9838
0.20	0.2227	0.80	0.7421	1.8	0.9891
0.25	0.2763	0.85	0.7707	1.9	0.9928
0.30	0.3286	0.90	0.7970	2.0	0.9953
0.35	0.3794	0.95	0.8209	2.2	0.9981
0.40	0.4284	1.00	0.8427	2.4	0.9993
0.45	0.4755	1.1	0.8802	2.6	0.9998
0.50	0.5205	1.2	0.9103	2.8	0.9999

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